Unit IV: Plant Anatomy (Structural Organisation)

Chapter 9

Tissue and Tissue System

Learning Objectives

The learner will be able to,

- *Study major types of plant tissues and their functions.*
- *Describes the various type of tissue system*
- *Interpret cross sections and longitudinal sections of dicot and monocot root, stem and leaf.*
- *Compare the internal organization of dicot root and monocot root.*

Chapter Outline

9.1 Meristematic tissue
9.2 Permanent tissues
9.3 The tissue system
9.4 Epidermal tissue system
9.5 Fundamental tissue system
9.6 Vascular tissue system
9.7 Comparision of primary structure

This chapter introduces the internal structure of higher plants. The study of internal structure and organisation of plant is called Plant Anatomy (*Gk: Ana = as under; temnein = to cut*). Plants have cells as the basic unit. The cells are organised into tissues. The tissues in turn are organised into organs. The different organs in a plant have different internal structures. It is studied by means of dissection and microscopic examination.

Nehemiah Grew
Father of Plant Anatomy



1641–1712

Katherine Esau (1898–1997)

A legendary Role model for women in science. She was a scintillating Botany teacher and pioneering researcher for six decades. Her classic book **Anatomy of Seed Plants** is the best literature in Plant Anatomy. In recognition of her distinguished service to science, she was awarded **National Medal of Science (1989) by USA.**



The Tissues

A Tissue is a group of cells that are alike in origin, structure and function. The study of tissues is called Histology. A plant is made up of different types of tissues.

There are two principal groups:

1. Meristematic tissues
2. Permanent tissues

9.1 Meristematic Tissue

9.1.1 Characteristics and classification

The characters of meristematic tissues: (Gr. *Meristos*-**Divisible**)

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